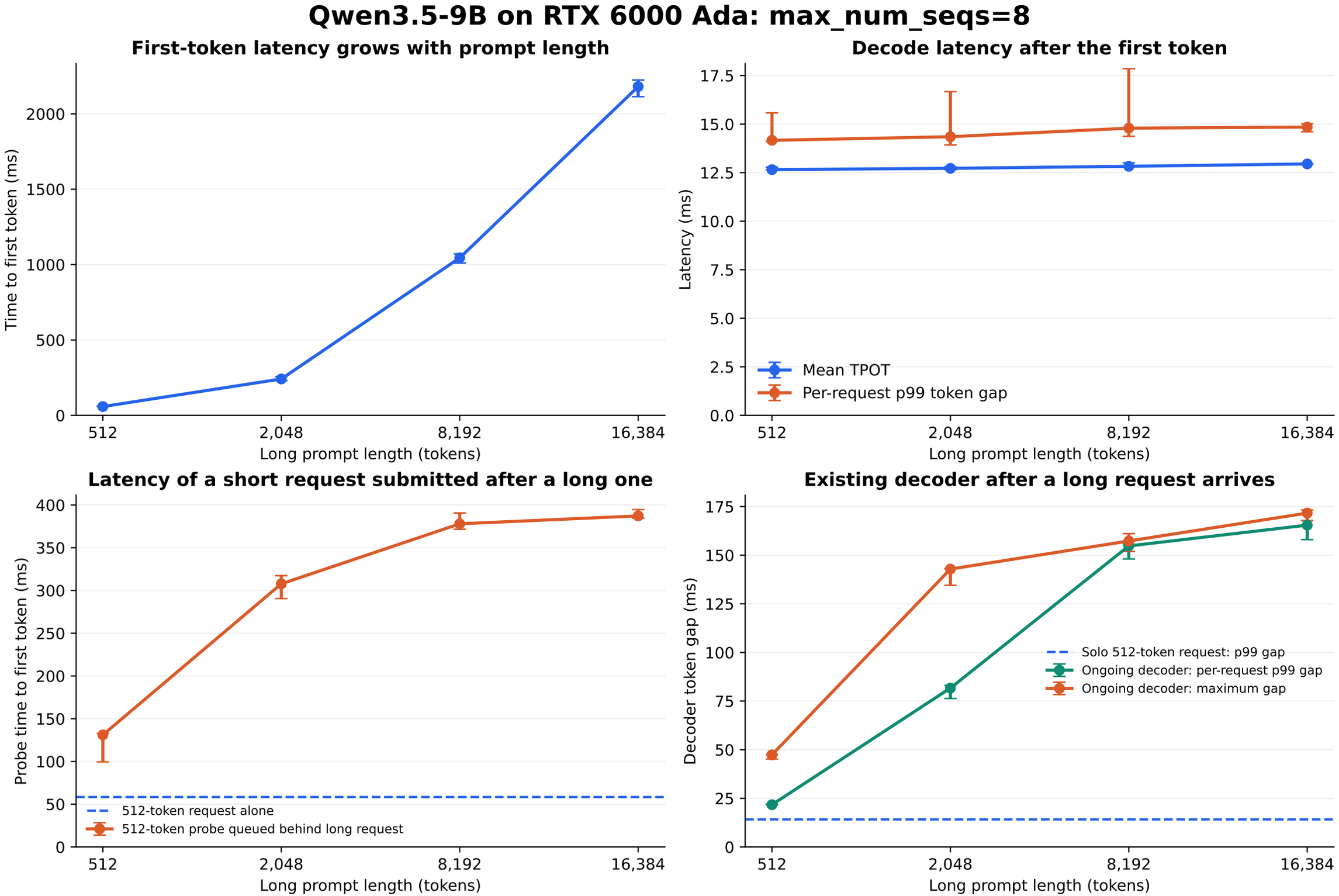
Qwen3.5-9B on RTX 6000 Ada: max_num_seqs=8
First-token latency grows with prompt length
Time to first token (ms)
0
500
1000
1500
2000
512
2,048
8,192
16,384
Long prompt length (tokens)
Decode latency after the first token
Latency (ms)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Mean TPOT
Per-request p99 token gap
512
2,048
8,192
16,384
Long prompt length (tokens)
Latency of a short request submitted after a long one
Probe time to first token (ms)
0
50
100
150
200
250
300
350
400
512-token request alone
512-token probe queued behind long request
512
2,048
8,192
16,384
Long prompt length (tokens)
Existing decoder after a long request arrives
Decoder token gap (ms)
0
25
50
75
100
125
150
175
Solo 512-token request: p99 gap
Ongoing decoder: per-request p99 gap
Ongoing decoder: maximum gap
512
2,048
8,192
16,384
Long prompt length (tokens)
GPTQ INT4 weights · FP8 KV · eager · max_num_seqs=8 · 3 repetitions · median with min–max bars
Cold-prefix requests; loopback client timings. Concurrent prefill/decode; single GPU; no KV transfer.